

An Audited Computational Framework for Knot-Invariant Inequalities

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Two Ways a Database Search Fails

Motivation. Mining a knot-invariant database (KnotInfo, 12,967 knots) for inequalities $A(K) \leq B(K)$ is cheap and tempting. It fails two ways that are easy to miss:

- A routine can be **structurally incapable** of finding what it claims to test for.
- A bound can hold on **every tabulated example** and still be false beyond the database’s crossing-number ceiling.

This project hit both, caught both, and built validated tools in response.

Failure Mode 1: A Tautological Engine

An early signature routine for positive 3-braids returned $\sigma(\hat{\beta}) = -(e - 2)$ **for every input**, regardless of the braid word – restating the crossing number e , never actually computing a linking form. “Zero defects in 1,280 configurations” was forced by the code, not discovered.

Counterexample once fixed: $T(3, 7)$, a genuine positive 3-braid knot (14 crossings, braid index 3):

$$s = 12, \quad |\sigma| = 8, \quad \text{defect} = 4.$$

Re-run with a validated engine, **84–89%** of knots in two independent 3-braid families show a nonzero defect, at crossings as low as 10.

Failure Mode 2: True on Every Tabulated Knot, Still False

The bound $g_T(K) \leq \text{braid index}(K) - 1$ held with **zero violations across all 2,953** KnotInfo knots with both invariants tabulated.

Checked against the literature (Abe–Kishimoto 2010; Lowrance 2011) before reporting it:

$$g_T(T_{3,3n+i}) = n \quad (n \geq 0, i \in \{0, 1, 2\})$$

Braid index of $T(3, q)$ is fixed at 3 for $q > 3$; g_T grows without bound. **First failure:** $T(3, 10)$, 20 crossings, $g_T = 3 > 2$.

KnotInfo stops at 12 crossings. The counterexample is 8 crossings past the ceiling — an exhaustive database sweep could never have found it.

Also Found: Six More Silent Bugs

Separate from the two failures above, six `print()` statements across four analysis scripts contained hardcoded narrative claims that did not match the data computed two lines above them in the same script — e.g. one asserted defect-knot determinants “skew towards 1 and 7 mod 8” when the actual dominant residues are 5 and 3 (70.8% combined, vs. 29% for 1 and 7). All six were traced, fixed to derive their claim from the real computation, and re-run.

Two Validated, From-Scratch Engines

Seifert-matrix signature (Collins’ 2007 algorithm, positive braids): homology generators between consecutive same-index crossings, linking numbers by a 5-case rule. Validated against 34 torus knots (closed-form signature) and all 17 KnotInfo knots with an authentic positive-braid word — **0 mismatches**.

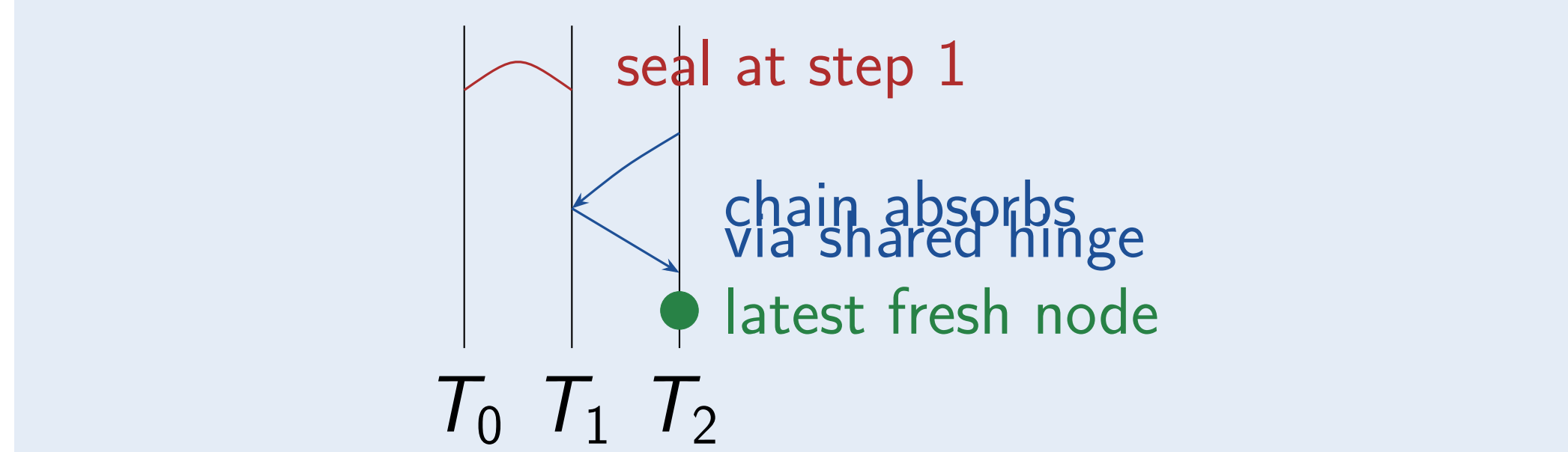
Turaev genus of a diagram (Kauffman-state / Temperley–Lieb circle counting, union-find implementation): validated against 3 alternating knots (must give exactly $g_T=0$) and **596 real KnotInfo knots** — never below the true tabulated value (mathematically required), exactly tight in 25.3% of cases.

The Hinge Mechanism (Proof Sketch)

Theorem. For a positive braid word on 3 strands with k syllables (maximal same-generator blocks) and length e :

$$g_T(D) = \left\lfloor \frac{k}{2} \right\rfloor - 1, \quad k \geq 2,$$

independent of e and of how crossings are split among syllables.



Lemma 1 (reduction). Repeats within a syllable spin off isolated circles; only one crossing per syllable matters. Reduces to k -length pure alternation.

Lemma 2 (hinge induction). $\sigma_1=\text{cap}(0, 1)$ and $\sigma_2=\text{cap}(1, 2)$ share strand 1. Before closure: exactly 3 components always — $\{T_0, T_1\}$ sealed at step 1, a chain absorbing T_2 one node per step, and the newest fresh node. Closure parity gives 2 components (k odd) or 1 (k even).

Zero mismatches: base case $k=1..40$, 160 random distributions, 203 further (e, k) pairs.

Why 8_{19} Motivated This

KnotInfo’s own word for $8_{19} = T(3, 4)$, $[1, 1, 1, 2, 1, 1, 1, 2]$ ($k=4$ syllables), gives $g_T(D) = 1$ — the true minimum. The naive torus word $[1, 2, 1, 2, 1, 2, 1, 2]$ ($k=8$) for the *same knot, same crossing number, same braid index* gives $g_T(D) = 3$. Syllable count, not crossing number, controls diagram genus.

Correction: Not Original — Caught the Same Way

Checked against literature before reporting. It failed.

A. Lowrance (*Comment. Math. Helv.* 86, 2011) already proves the reduction lemma in general (**Corollary 3.10**, any braid index), building on Champanerkar–Kofman. His Proposition 4.3 even uses the even- k base case as an intermediate step.

What survives: an independent proof by a different technique (union-find/Temperley–Lieb vs. his tangle-twisting argument), the explicit closed form, and the validated reusable engine.

This is the **third** time in this project that checking a claim against data or literature before reporting it changed the conclusion.

Beyond 3 Strands: Labeled a Conjecture, Not a Theorem

Cyclic alternation for $n = 3..9$ strands:

$$\text{period} = \begin{cases} 2 & n-1 \text{ even} \\ n-1 & n-1 \text{ odd} \end{cases}$$

$$\text{growth per syllable} = \frac{\lfloor (n-1)/2 \rfloor}{n-1}$$

Consistent with 2-colorability of the generator cycle C_{n-1} — not proved. **No literature check done yet; no proof attempted for $n \geq 5$.** Reported honestly as future work, not a result.

Results at a Glance

Claim	Status	Evidence
Positive 3-braids signature-rigid	Retracted	tautological engine
$g_T \leq \text{braid index} - 1$	Refuted	$T(3, 10)$ + cited thm.
Syllable formula ($n=3$)	Re-proved	Lowrance 2011 (prior)
Syllable formula ($n \geq 4$)	Conjecture	computation only

Discussion & Future Work

The most reliable output of this project was not a single inequality but a **discipline**: re-run from a clean checkout and diff against the claim; check any surviving claim against the literature before reporting it. Applied consistently, it changed the conclusion three separate times.

Next: literature search for the $n \geq 4$ conjecture; a proof via the hinge structure of the generator cycle C_{n-1} ; extend both engines to mixed-sign (non-positive) braid words.

Full reproducible record: `manuscript/log.tex` (20 dated entries), every number traceable to a script and a saved file under `results/`. Paper: `manuscript/isef_paper.tex`.